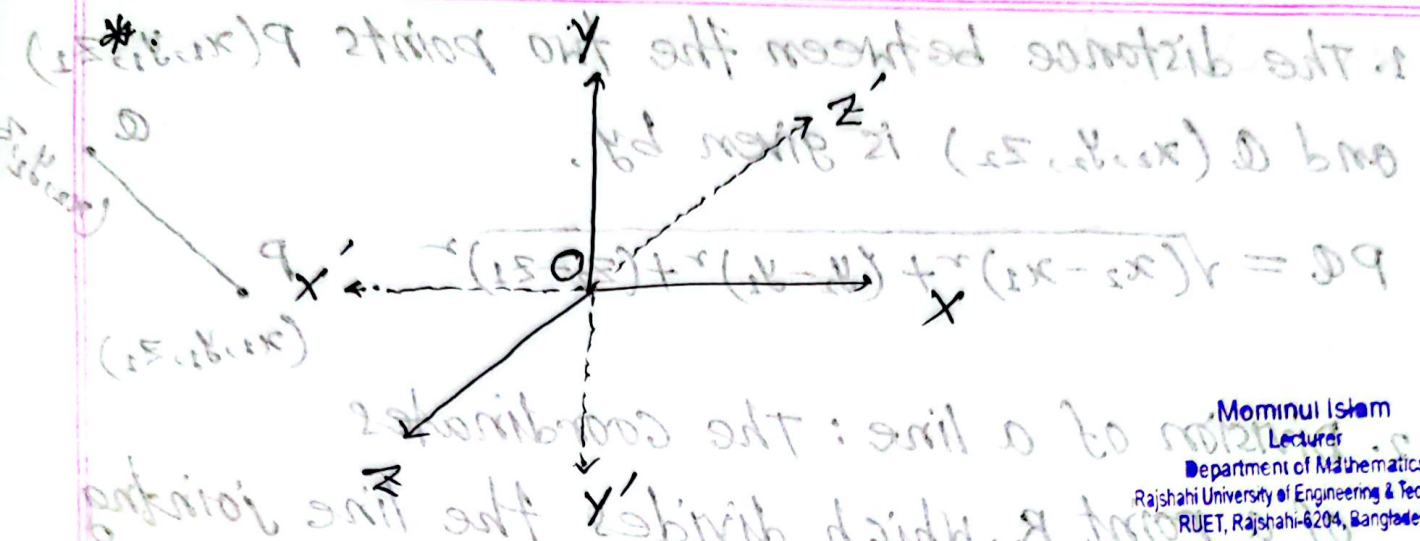
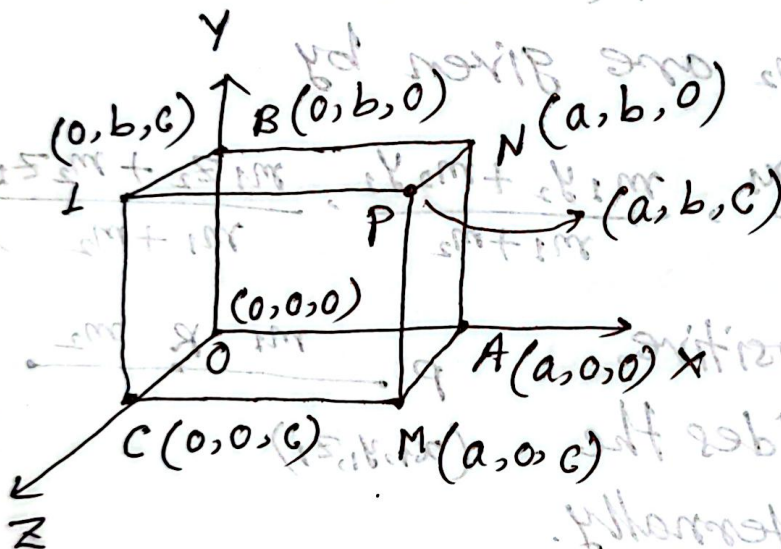


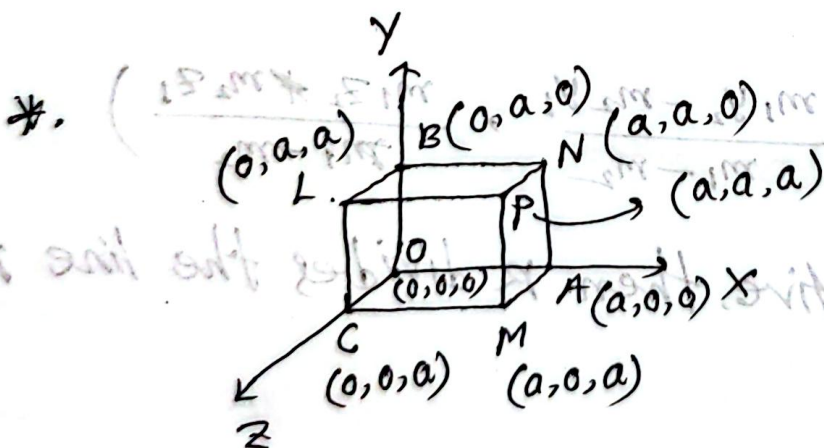
Rectangular Co-ordinates



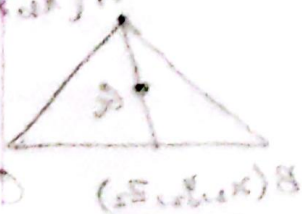
Mominul Islam
Lecturer
Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh



Rectangular
parallel pipe
be a, b, c .

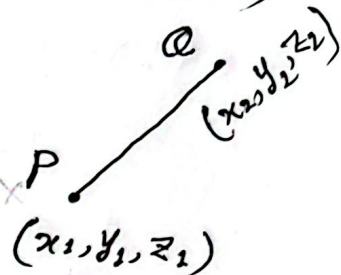


'a' be the
length of each
side of a cube.



1. The distance between the two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by,

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

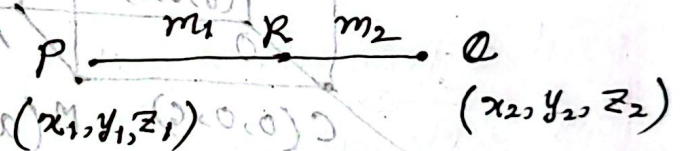


2. Division of a line: The coordinates of a point R , which divides the line joining the two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio $m_1 : m_2$ are given by

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}, \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2} \right)$$

If $\frac{m_1}{m_2}$ is positive,

then R divides the line PQ internally.

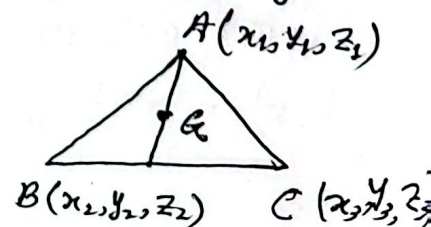


$$\left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}, \frac{m_1 z_2 - m_2 z_1}{m_1 - m_2} \right)$$

If $\frac{m_1}{m_2}$ is negative, then R divides the line PQ externally.

3. The co-ordinates of the centre of gravity of a $\triangle ABC$ are

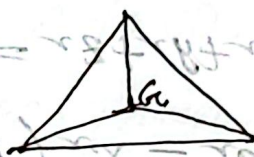
$$G \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$



4. The centre of gravity of a tetrahedron the co-ordinates of whose vertices are (x_r, y_r, z_r) where $r=1, 2, 3, 4$ is the point G .

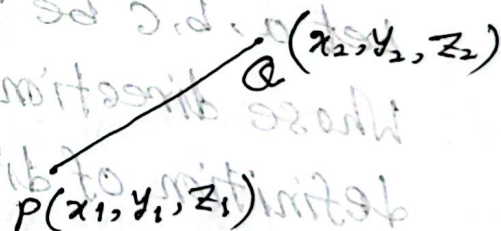
$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

Mominul Islam
Lecturer
Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6100.



5. Direction ratios: Any three numbers a, b and c which are proportional to the direction cosines l, m and n respectively of a given line are called the direction ratios of the given line.

$$a \propto l, b \propto m, c \propto n.$$



Here, the direction ratios of the line PQ are

$$x_2 - x_1, y_2 - y_1, z_2 - z_1.$$

6. Direction cosines: If any line makes angle α, β, γ on with the positive direction of the axes x, y, z respectively then the $\cos \alpha, \cos \beta, \cos \gamma$ are called the direction cosines of the given line.

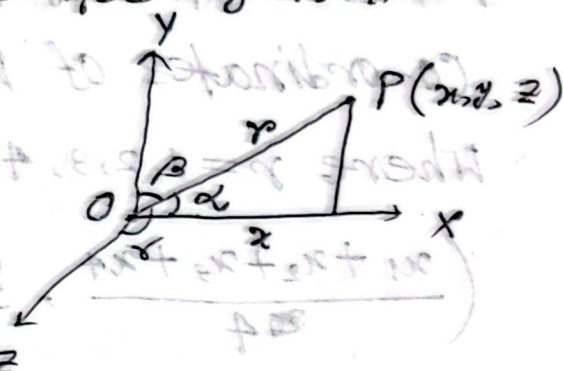
Direction cosines are represented by l, m, n .

(= Here, $\cos \alpha = \frac{x}{r}$

$\Rightarrow x = r \cos \alpha = lr$

similarly, $y = r \cos \beta = mr$

$z = r \cos \gamma = nr$



$\therefore x + y + z = r(l + m + n)$

$(l + m + n)r = r$

$\therefore l + m + n = 1$

7. Relation between direction cosines and direction ratios:

Let a, b, c be the direction ratios of a line whose direction cosines are l, m, n . Now from definition of direction ratios, we have

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}} = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}$$

$$\therefore l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

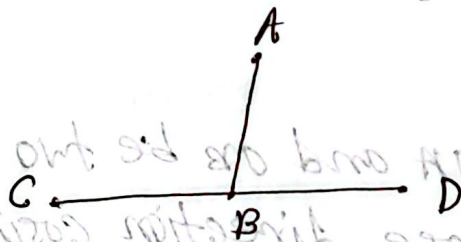
Where the same sign either (+) axis or (-) axis is to be chosen throughout.

Hence if a, b, c be three numbers proportional to the direction cosines of OP, then the actual

Direction cosines of OP are $\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$.

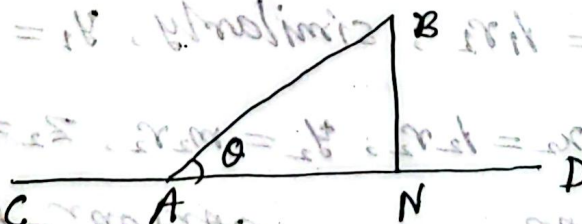
and that of PO are $\frac{-a}{\sqrt{a^2+b^2+c^2}}, \frac{-b}{\sqrt{a^2+b^2+c^2}}, \frac{-c}{\sqrt{a^2+b^2+c^2}}$.

8. Projection of a point on a straight line: Let A be a given point and CD the given straight line. Draw perpendicular AB from A to CD , where B is the foot of the perpendicular. Then the foot of the perpendicular B is called the projection of the given point A on the given line CD .



9. Projection of a line on another straight line:

The projection of a given segment AB of a line on any line CD is $AB \cos \theta$, where θ being the angle between the two lines AB and CD .



Mominul Islam
Lecturer

Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh

10. The ~~proj~~ection of a line joining the two points (x_1, y_1, z_1) and (x_2, y_2, z_2) on a straight line whose direction cosines are l, m, n is

$$l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

Q.1. If θ be the angle between the two straight lines whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 then show that,

$$i. \cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

Also find the condition of parallel and perpendicular.

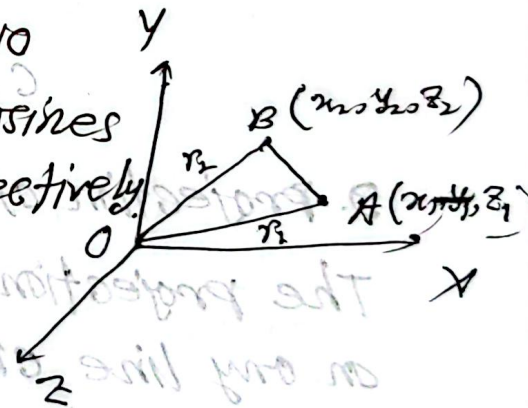
Solution: Let, OA and OB be two given lines whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 respectively.

$$\begin{aligned} \text{Now, } \frac{l_1}{x_1} &= \frac{m_1}{y_1} = \frac{n_1}{z_1} \\ &= \frac{\sqrt{l_1^2 + m_1^2 + n_1^2}}{\sqrt{x_1^2 + y_1^2 + z_1^2}} = \frac{1}{OA} \end{aligned}$$

$$\therefore x_1 = l_1 r_1; \text{ similarly, } y_1 = m_1 r_1, z_1 = n_1 r_1$$

$$\text{Again, } x_2 = l_2 r_2, y_2 = m_2 r_2, z_2 = n_2 r_2$$

$$\text{From } \triangle OAB, \cos \theta = \frac{OA^2 + OB^2 - AB^2}{2 \cdot OA \cdot OB}$$



$$\Rightarrow \cos \theta = \frac{(x_1^2 + y_1^2 + z_1^2) + (x_2^2 + y_2^2 + z_2^2) - \{ (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \}}{2 \cdot \vec{OA} \cdot \vec{OB}}$$

$$= \frac{(1^2 r_1^2 + m_1^2 r_1^2 + n_1^2 r_1^2) + (1^2 r_2^2 + m_2^2 r_2^2 + n_2^2 r_2^2) - \{ (1^2 r_2^2 - 1^2 r_1^2) + (m_2^2 r_2^2 - m_1^2 r_1^2) + (n_2^2 r_2^2 - n_1^2 r_1^2) \}}{2 r_1 r_2}$$

$$\therefore \cos \theta = 1/1 + m_1 m_2 + n_1 n_2$$

$$\text{When } \theta = 90^\circ; \text{ then } 1/1 + m_1 m_2 + n_1 n_2 = 0$$

$$\text{When } \sin \theta = 1 - \cos^2 \theta$$

$$= (1^2 + m_1^2 + n_1^2)(1^2 + m_2^2 + n_2^2)$$

$$- (1/1 + m_1 m_2 + n_1 n_2)^2$$

$$= \{ (1^2 m_2 - 1^2 m_1)^2 + (m_1 m_2 - m_2 n_1)^2 + (n_1 n_2 + n_2 n_1)^2 \}$$

$$\theta = 0^\circ, \text{ then } \frac{1_1}{1_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} \quad \underline{\underline{Ans}}$$

$$** \cos \theta = 1/1 + m_1 m_2 + n_1 n_2$$

$$= \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$1_1 = \frac{a_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}}$$

$$1_2 = \frac{a_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Q.2. Find the direction cosines of the line drawn from the origin to the point $(-6, 2, 3)$.

Solution: The direction ratios of the line joining the two points $O(0, 0, 0)$ and $P(-6, 2, 3)$ are

$$(-6 - 0, 2 - 0, 3 - 0)$$

$$= -6, 2, 3$$

$$\text{Now, } OP = \sqrt{(-6)^2 + 2^2 + 3^2} = 7.$$

$$\therefore l = \frac{-6}{7}, m = \frac{2}{7}, n = \frac{3}{7}$$

Hence the required direction cosines are

$$-\frac{6}{7}, \frac{2}{7}, \frac{3}{7}.$$

Q.3. Find the direction cosines of the line which is equally inclined to the axes.

Solution: Let, l, m, n be the required direction cosines. Thus as the line makes equal angles with the axes, so we have, $\alpha = \beta = \gamma$

$$\Rightarrow \cos \alpha = \cos \beta = \cos \gamma$$

$$\Rightarrow l = m = n \rightarrow (1)$$

$$\text{We know, } l^2 + m^2 + n^2 = 1$$

$$\Rightarrow l^2 + l^2 + l^2 = 1 \Rightarrow l = \pm \frac{1}{\sqrt{3}}, m = \pm \frac{1}{\sqrt{3}}, n = \pm \frac{1}{\sqrt{3}}$$

The required direction cosines are

$$\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}.$$

Q. 4 Find the distance of $(-1, 2, 5)$ from the line through $(3, 4, 5)$ whose direction cosines are proportional to $2, -3, 6$.

Solution: Let BC be the line passing through the point $B(3, 4, 5)$ whose direction cosines are proportional to $2, -3, 6$.

Let l, m, n be the direction cosine of BC.

$$\therefore l = \frac{2}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{2}{7}$$

$$\text{Similarly, } m = -\frac{3}{7}, n = \frac{6}{7}$$

$$AB = \sqrt{(3+1)^2 + (4-2)^2 + (5-5)^2} = \sqrt{20} = 2\sqrt{5}$$

Draw AD from A perpendicular to BC.

BD = The projection of BA on BC

$$= l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

$$= \frac{2}{7}(-1-3) - \frac{3}{7}(2-4) + \frac{6}{7}(5-5)$$

$$= -\frac{2}{7}$$

From the right angle triangle ABD, $AD = \sqrt{AB^2 - BD^2}$

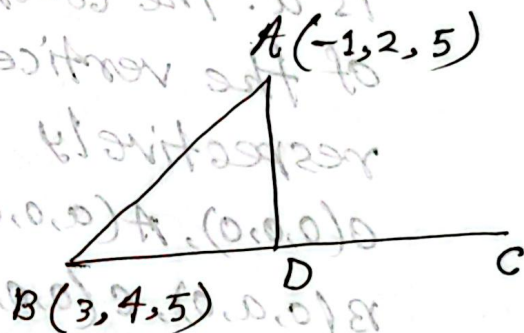
$$= \sqrt{20 - \frac{4}{49}}$$

$$= \sqrt{\frac{976}{49}}$$

$$= \frac{4\sqrt{61}}{7} \text{ Ans.}$$

Mominul Islam
Lecturer

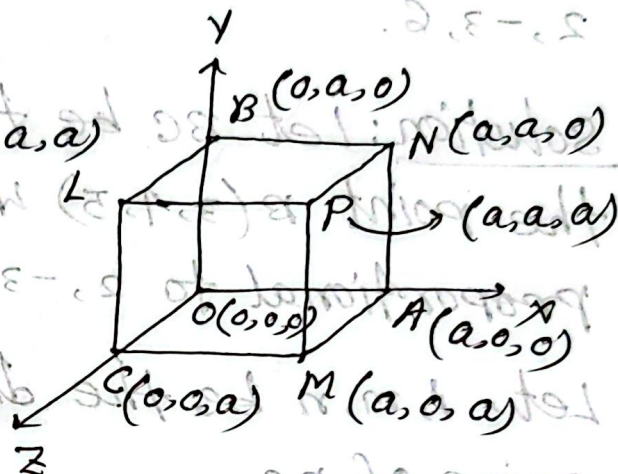
Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh



Q.5. prove that the angle between two diagonals of a cube is $\cos^{-1}(\frac{1}{3})$.

Solution:

Let OCMANBLP be a cube whose edge is a . The co-ordinates of the vertices are respectively



$O(0,0,0), A(a,0,0),$

$B(0,a,0), C(0,0,a), M(a,0,a), N(a,a,0), L(0,a,a), P(a,a,a).$

The direction ratios of the diagonals OP are $a-0, a-0, a-0$

i.e., a, a, a

The direction ratios of the diagonals CN are

$a-0, a-0, 0-a$

i.e., $a, a, -a$

Let, θ be the angle between OP and CN

$$\begin{aligned} \therefore \cos \theta &= \frac{a \cdot a + a \cdot a + a \cdot (-a)}{\sqrt{a^2 + a^2 + a^2} \cdot \sqrt{a^2 + a^2 + a^2}} \\ &= \frac{a^2}{\sqrt{3a^2} \cdot \sqrt{3a^2}} \\ &= \frac{1}{3} \end{aligned}$$

$$\therefore \theta = \cos^{-1}\left(\frac{1}{3}\right). \text{ (proved)}$$

Q.6. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube prove that,

i. $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$

ii. $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta = \frac{8}{3}$

Solution: i. Let OGMANBLP be a cube whose edge is a .

The direction cosines of the diagonal OP are

$$\frac{a-0}{\sqrt{a^2+a^2+a^2}}, \frac{a-0}{\sqrt{a^2+a^2+a^2}}, \frac{a-0}{\sqrt{a^2+a^2+a^2}}$$

i.e., $\frac{a}{\sqrt{3}a}, \frac{a}{\sqrt{3}a}, \frac{a}{\sqrt{3}a}$

i.e., $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Similarly, the direction cosines of the diagonal AL

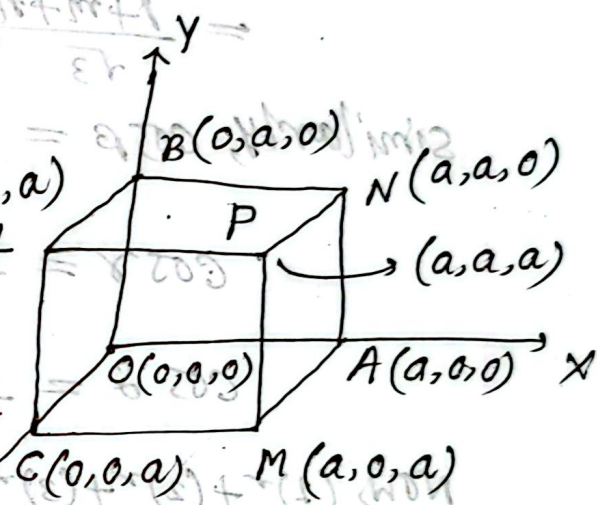
are, $\frac{0-a}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a}, \frac{a-0}{\sqrt{3}a} = -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The direction cosines of the diagonal BM are

$\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

The direction cosines of the diagonal CN are

$\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$



Mominul Islam
Lecturer
Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh

Let l, m, n be direction cosines of the line which makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals OP, AL, BM and CN respectively,

$$\therefore \cos \alpha = l \left(\frac{1}{\sqrt{3}} \right) + m \left(\frac{1}{\sqrt{3}} \right) + n \left(\frac{1}{\sqrt{3}} \right)$$

$$= \frac{l+m+n}{\sqrt{3}} \rightarrow (1)$$

$$\text{Similarly, } \cos \beta = \frac{-l+m+n}{\sqrt{3}} \rightarrow (2)$$

$$\cos \gamma = \frac{l-m+n}{\sqrt{3}} \rightarrow (3)$$

$$\cos \delta = \frac{l+m-n}{\sqrt{3}} \rightarrow (4)$$

$$\text{Now, } (1)^2 + (2)^2 + (3)^2 + (4)^2 =$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \left(\frac{l+m+n}{\sqrt{3}} \right)^2$$

$$+ \left(\frac{-l+m+n}{\sqrt{3}} \right)^2 + \left(\frac{l-m+n}{\sqrt{3}} \right)^2 + \left(\frac{l+m-n}{\sqrt{3}} \right)^2$$

$$= \frac{1}{3} \cdot 4(l^2 + m^2 + n^2)$$

$$= \frac{4}{3} \cdot (\text{proved})$$

$$\text{ii. } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

$$\Rightarrow 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma + 1 - \sin^2 \delta = \frac{4}{3}$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + \sin^2 \delta = 4 - \frac{4}{3}$$

$$= \frac{8}{3} \cdot (\text{proved})$$

7. If the direction cosines of two lines are connected by the relation $2l+2m-n=0$ and $lm+mn+nl=0$ then find the direction cosines of the lines. show that the lines are perpendicular to each other.

Solution: Given that, $2l+2m-n=0 \rightarrow (1)$

$$\& \quad lm+mn+nl=0 \rightarrow (2)$$

$$\text{From (1)} \Rightarrow n=2l+2m \rightarrow (3)$$

Using (3), from (2) we get,

$$lm+m(2l+2m)+(2l+2m)l=0$$

$$\Rightarrow 2l^2+5lm+2m^2=0$$

$$\Rightarrow (2l+m)(2m+l)=0$$

$$\therefore 2l+m=0 \rightarrow (4)$$

$$\& \quad 2m+l=0 \rightarrow (5)$$

From (1) & (4), we get

$$\frac{l}{+1} = \frac{m}{-2} = \frac{n}{-2}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{2} = \frac{n}{2} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{1+4+4}}$$

$$\therefore l = -\frac{1}{3}, m = \frac{2}{3}, n = \frac{2}{3}$$

Similarly, from (1) & (5), we get,

$$\frac{l}{2} = \frac{m}{-1} = \frac{n}{2} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{4+1+4}} = \frac{1}{3}$$

$$\therefore l = \frac{2}{3}, m = -\frac{1}{3}, n = \frac{2}{3}$$

The two lines are perpendicular of

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\text{NOW, } -\frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{-1}{3} + \frac{2}{3} \cdot \frac{2}{3}$$

$$= \frac{-2-2+4}{9}$$

$$= 0.$$

Hence the given lines are perpendicular each other. (shoved)

8. If the direction cosines of two lines are connected by the following relations then find the angle between the lines

i. $l+m+n=0$ and $l^2+m^2-n^2=0$

ii. $l-5m+3n=0$ and $7l^2+5m^2-3n^2=0$

iii. $l+m+n=0$ and $2lm+2ln-mn=0$

i. solution: Given that, $l+m+n=0 \rightarrow (1)$

$$\& \quad l^2+m^2-n^2=0 \rightarrow (2)$$

$$\text{From (1)} \Rightarrow n = -l-m \rightarrow (3)$$

Using (3), from (2) we get

$$l^2+m^2-(l+m)^2=0$$

$$\Rightarrow 2lm=0$$

$$\Rightarrow lm=0$$

$$\therefore l=0$$

$$\Rightarrow l+0+0=0 \rightarrow (4) \quad \left| \begin{array}{l} \text{or, } m=0 \\ \Rightarrow 0+m+n=0 \rightarrow (5) \end{array} \right.$$

From (1) & (4) we get

$$\frac{l}{0} = \frac{m}{1} = \frac{n}{-1} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{0^2+1^2+1^2}} = \frac{1}{\sqrt{2}}$$

$$\therefore l = 0, m = \frac{1}{\sqrt{2}}, n = -\frac{1}{\sqrt{2}}$$

From (1) & (5), we get

$$\frac{l}{-1} = \frac{m}{0} = \frac{n}{-1} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{1^2+0^2+1^2}} = \frac{1}{\sqrt{2}}$$

$$\therefore l = -\frac{1}{\sqrt{2}}, m = 0, n = -\frac{1}{\sqrt{2}}$$

If θ be the angle between them, then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$= 0 \cdot \left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} \cdot 0 + \left(-\frac{1}{\sqrt{2}}\right) \cdot \left(-\frac{1}{\sqrt{2}}\right)$$

$$= \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3} \quad \underline{\text{Ans.}}$$

Mominul Islam,

Lecturer

Department of Mathematics

Rajshahi University of Engineering & Technology

RUET, Rajshahi-6204, Bangladesh

ii. iii. solution : ———, same as.

9. Two straight lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$. prove that the lines are perpendicular if $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$ and the lines are parallel if $\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$.

Solution: The given relations are,

$$al + bm + cn = 0 \longrightarrow (1)$$

$$\& fm + gn + h/m = 0 \longrightarrow (2)$$

$$\text{From (1)} \Rightarrow n = \frac{-al - bm}{c} \longrightarrow (3)$$

using (3), from (2), we get

$$fm \left(\frac{-al - bm}{c} \right) + gn \left(\frac{-al - bm}{c} \right) + h/m = 0$$

$$\Rightarrow -af/m - bfm - ag/r - bg/m + ch/m = 0$$

$$\Rightarrow ag/r + (af + bg - ch)/m + bfm = 0 \longrightarrow (4)$$

$$\Rightarrow ag \left(\frac{1}{m} \right)^2 + (af + bg - ch) \cdot \left(\frac{1}{m} \right) + bf = 0 \longrightarrow (5)$$

Equation (5) is a quadratic equation of $\frac{1}{m}$. So it has two roots. Let, $\frac{r_1}{m_1}$ and $\frac{r_2}{m_2}$ be the two roots of the equation.

$$\therefore \frac{r_1}{m_1} \cdot \frac{r_2}{m_2} = \frac{bf}{ag} = \frac{f/a}{g/b}$$

$$\Rightarrow \frac{r_1 r_2}{f/a} = \frac{m_1 m_2}{g/b} \longrightarrow (6)$$

similarly, using from (1) & (2) eliminating l ,

$$\text{we get, } \frac{m_1 m_2}{g/b} = \frac{n_1 n_2}{h/c} \longrightarrow (7)$$

From (6) & (7), we get,

$$\frac{r_1 r_2}{f/a} = \frac{m_1 m_2}{g/b} = \frac{n_1 n_2}{h/c} = k$$

$$\therefore l_1/l_2 = k \cdot f/a, m_1/m_2 = k \cdot g/b, n_1/n_2 = k \cdot h/c$$

$$\text{Now, } l_1 l_2 + m_1 m_2 + n_1 n_2 = k(f/a + g/b + h/c)$$

The two lines will be perpendicular if

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\Rightarrow k(f/a + g/b + h/c) = 0$$

$$\therefore f/a + g/b + h/c = 0$$

Again, the two lines will be parallel if

$$\frac{l_1}{l_2} = \frac{m_1}{m_2} \Rightarrow \frac{l_1}{m_1} = \frac{l_2}{m_2}$$

i.e., the two roots of equation (3) are equal.

$$(af + bg - ch)^2 - 4 \cdot ag \cdot bf = 0$$

$$\Rightarrow af + bg - ch = \pm 2\sqrt{ag \cdot bf}$$

$$\Rightarrow af \pm 2\sqrt{ag \cdot bf} + bg = ch$$

$$\Rightarrow (\sqrt{af} \pm \sqrt{bg})^2 = ch$$

$$\Rightarrow \sqrt{af} \pm \sqrt{bg} = \pm \sqrt{ch}$$

$$\Rightarrow \sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0 \quad (\text{proved})$$

Mominul Islam
Lecturer

Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh

10. Prove that the two straight lines whose direction cosines are given by the two relations $al + bm + cn = 0$ and $u^2 + v^2 + w^2 = 0$ are perpendicular or parallel according as $a^2(v+w) + b^2(w+u) + c^2(u+v) = 0$

$$\text{or } \frac{ar}{u} + \frac{br}{v} + \frac{cr}{w} = 0.$$

Solution: the given two relations are

$$ar + bm + cn = 0 \longrightarrow (1)$$

$$\& ur^2 + vm^2 + wn^2 = 0 \longrightarrow (2)$$

$$\text{From (1)} \Rightarrow n = - \left(\frac{ar + bm}{c} \right) \longrightarrow (3)$$

Using (3), from (2) $\Rightarrow$

$$ur^2 + vm^2 + w \left\{ - \left(\frac{ar + bm}{c} \right) \right\}^2 = 0$$

$$\Rightarrow ucr^2 + vcm^2 + w(ar^2 + 2abrm + b^2m^2) = 0$$

$$\Rightarrow (ucr + war)r^2 + 2wabrm + (wb^2 + vcr)m^2 = 0$$

$$\Rightarrow (ucr + war) \left(\frac{r}{m} \right)^2 + 2wab \left(\frac{r}{m} \right) + (wb^2 + vcr) = 0$$

Let, r_1, m_1, n_1 and r_2, m_2, n_2 be the direction $\longrightarrow$ (4)
cosines of the given two lines.

Equation (4) is a quadratic equation of $\frac{r}{m}$. So it has two roots. Let $\frac{r_1}{m_1}$ and $\frac{r_2}{m_2}$ be the two roots of equation (4).

$$\therefore \frac{r_1}{m_1} \cdot \frac{r_2}{m_2} = \frac{wb^2 + vcr}{ucr + war}$$

$$\Rightarrow \frac{r_1 r_2}{wb^2 + vcr} = \frac{m_1 m_2}{ucr + war} \longrightarrow (5)$$

Similarly eliminating r from the relation (1) & (2), we get $\frac{m_1 m_2}{ucr + war} = \frac{n_1 n_2}{ar^2 + b^2u} \longrightarrow (6)$

From (5) & (6), we get

$$\frac{l_1 l_2}{ub^2 + vc^2} = \frac{m_1 m_2}{uc^2 + wa^2} = \frac{n_1 n_2}{av^2 + bu^2} = k$$

$$\therefore l_1 l_2 = k \cdot (ub^2 + vc^2), m_1 m_2 = k \cdot (uc^2 + wa^2), n_1 n_2 = k \cdot (av^2 + bu^2)$$

The two lines will be parallel if the two roots of equation (4) are equal. Hence the discriminant of equation (4) is equal to zero.

$$(2wab)^2 - 4(uc^2 + wa^2) \cdot (ub^2 + vc^2) = 0$$

$$\Rightarrow 4w^2 a^2 b^2 - 4uwb^2 c^2 - 4uvc^4 - 4w^2 a^2 b^2 - 4vw a^2 c^2 = 0$$

$$\Rightarrow uwb^2 c^2 + uvc^4 + vwa^2 c^2 = 0$$

$$\Rightarrow \frac{b^2}{v} + \frac{c^2}{u} + \frac{a^2}{w} = 0. \text{ (proved)}$$

Mominul Islam
Lecturer

Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh

11. If the edges of a rectangular parallelepiped are a, b, c . Show that the angles between the four diagonals are given by $\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$.

Solution: From the figure,

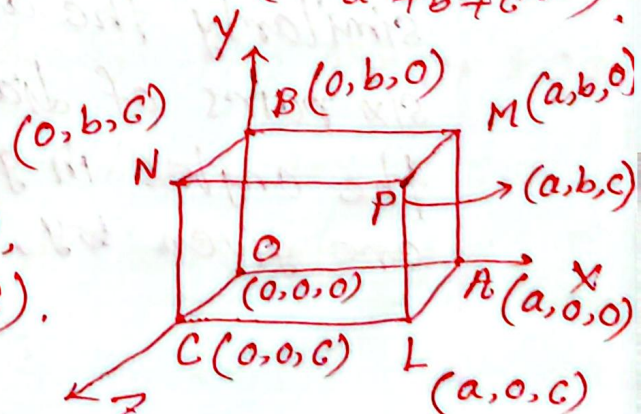
$$O(0,0,0), P(a,b,c),$$

$$A(a,0,0), B(0,b,0), C(0,0,c),$$

$$L(a,0,c), M(a,b,0), N(0,b,c).$$

Direction cosines of

$$OP \text{ are } \frac{a-0}{\sqrt{a^2+b^2+c^2}}, \frac{b-0}{\sqrt{a^2+b^2+c^2}}, \frac{c-0}{\sqrt{a^2+b^2+c^2}}.$$



$$= \frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

Direction cosines of CM are

$$\frac{a-0}{\sqrt{a^2+b^2+c^2}}, \frac{b-0}{\sqrt{a^2+b^2+c^2}}, \frac{0-c}{\sqrt{a^2+b^2+c^2}} = \frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{-c}{\sqrt{a^2+b^2+c^2}}$$

Direction cosines of AN are

$$\frac{0-a}{\sqrt{a^2+b^2+c^2}}, \frac{b-0}{\sqrt{a^2+b^2+c^2}}, \frac{c-0}{\sqrt{a^2+b^2+c^2}} = \frac{-a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

Direction cosines of BL are

$$\frac{a-0}{\sqrt{a^2+b^2+c^2}}, \frac{0-b}{\sqrt{a^2+b^2+c^2}}, \frac{c-0}{\sqrt{a^2+b^2+c^2}} = \frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{-b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

The angle between OP and CM,

$$\therefore \cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$= \frac{a \cdot a + b \cdot b + c \cdot (-c)}{\sqrt{a^2+b^2+c^2} \cdot \sqrt{a^2+b^2+c^2}} = \frac{a^2+b^2-c^2}{a^2+b^2+c^2}$$

$$\therefore \theta = \cos^{-1} \left(\frac{a^2+b^2-c^2}{a^2+b^2+c^2} \right)$$

similarly the angle between any one of the six pairs of diagonals can be found. Hence the angles in general between the six pairs are given by, $\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2} \right)$.

12. prove that $l^2 + m^2 + n^2 = 1$ Where l, m, n Direction cosines of any line.

Solution: Let $P(x, y, z)$ be the co-ordinates of any point on the line, $\cos \alpha = \frac{ON}{OP} = \frac{x}{r}$

$$\Rightarrow x = r \cos \alpha$$

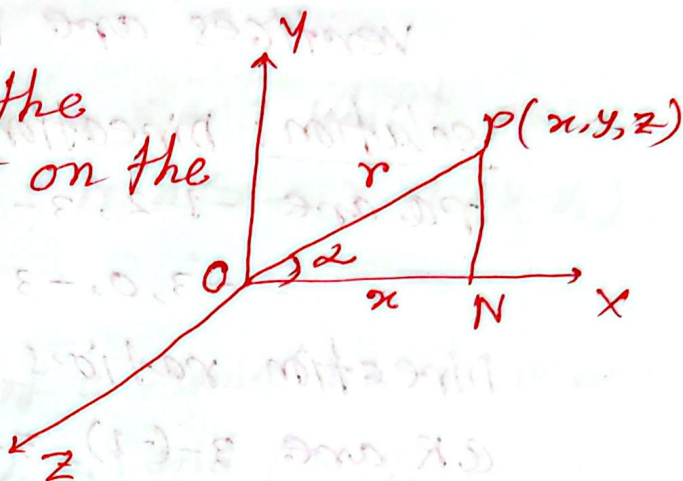
$$\Rightarrow x = r l \quad [\because l = \cos \alpha]$$

similarly, $y = r m$ and $z = r n$

$$\therefore x^2 + y^2 + z^2 = r^2 (l^2 + m^2 + n^2)$$

$$\Rightarrow r^2 = r^2 (l^2 + m^2 + n^2) \quad | \because r^2 = x^2 + y^2 + z^2$$

$$\therefore l^2 + m^2 + n^2 = 1. \text{ (proved)}$$



Mominul Islam

Lecturer

Department of Mathematics

Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh

13. Find the direction cosines of the straight line PQ, Where $P(2, -3, 4)$ & $Q(-1, 2, 3)$.

14. If the co-ordinates of P and Q are $(2, 3, 4)$ and $(1, 1, -1)$ respectively and O is the origin then find the angle between the line OP & OQ.

15. If direction ratios of two lines are $2, 1, 1$ and $4, \sqrt{3}-1, -\sqrt{3}-1$ then show that angle between the lines is $\pi/3$.

13, 14, 15 $\longrightarrow$ For practice

16. Find the angles of a triangle PQR whose vertices are $P(2, 3, 5)$, $Q(-1, 3, 2)$ & $R(3, 5, -2)$

solution: Direction ratios of PQ are $-1-2, 3-3, 2-5$

$$= -3, 0, -3$$

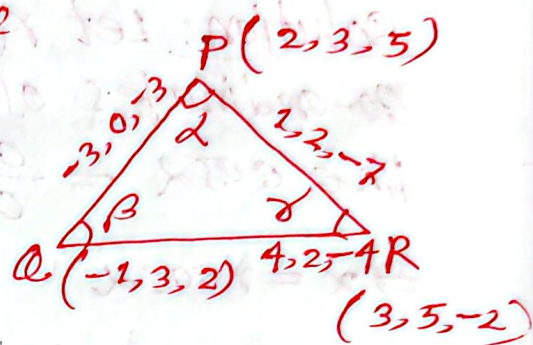
Direction ratios of

QR are $3-(-1), 5-3, -2-2$

$$= 4, 2, -4$$

Direction ratios of PR are $3-2, 5-3, -2-5$

$$= 1, 2, -7$$



$$\cos \alpha = \frac{-3 \cdot 1 + 0 \cdot 2 + (-3) \cdot (-7)}{\sqrt{(-3)^2 + 0^2 + (-3)^2} \cdot \sqrt{1^2 + 2^2 + (-7)^2}} = \frac{18}{\sqrt{18} \cdot \sqrt{54}}$$

$$\Rightarrow \alpha = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$\text{similarly, } \cos \beta = \frac{-3 \cdot 4 + 0 \cdot 2 + (-3) \cdot (-4)}{\sqrt{(-3)^2 + 0^2 + (-3)^2} \cdot \sqrt{4^2 + 2^2 + (-4)^2}} = 0$$

$$\therefore \beta = \frac{\pi}{2}$$

$$\text{Again, } \cos \gamma = \frac{4 \cdot 1 + 2 \cdot 2 + (-4) \cdot (-7)}{\sqrt{4^2 + 2^2 + (-4)^2} \cdot \sqrt{1^2 + 2^2 + (-7)^2}} = \frac{36}{6\sqrt{54}}$$

$$\therefore \gamma = \cos^{-1} \left(\sqrt{\frac{2}{3}} \right) \quad \underline{\underline{Ans.}}$$